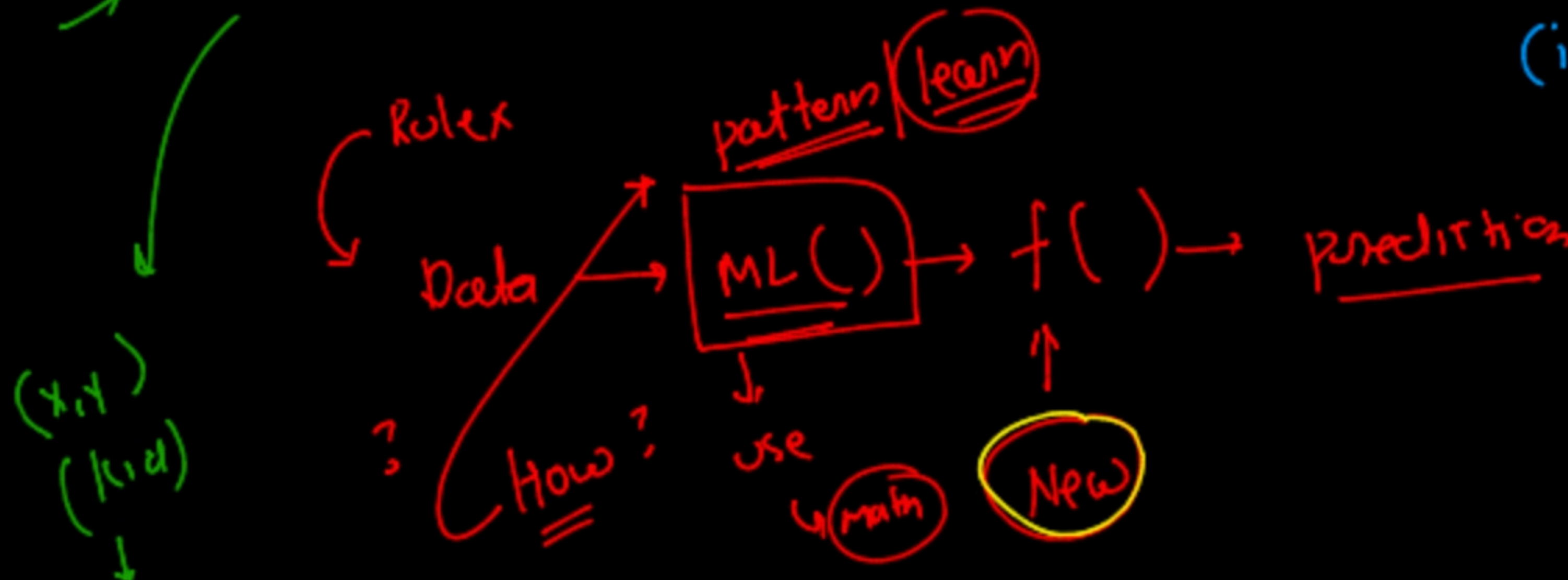
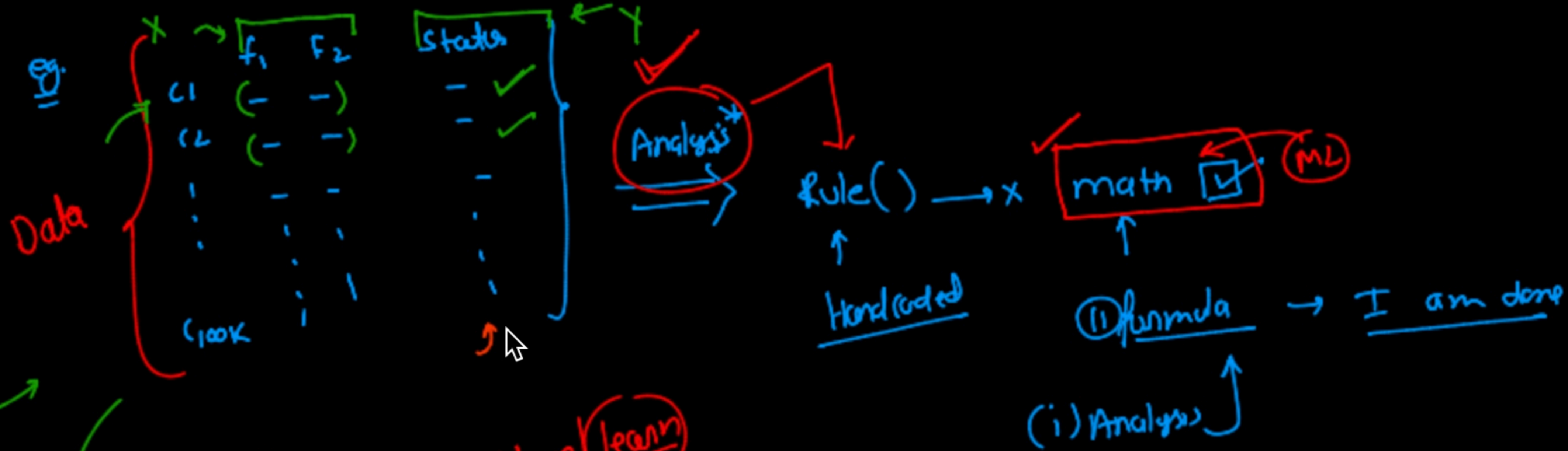




(x, y)
 (k, d)
 $\downarrow$
 $load() \leftarrow x$
 $\downarrow$
 $y \rightarrow E(A, Ne)$

Training pattern

$D(X, Y)$
↑
(labelled data) → G

Supervised training *

✓ (i) classification

Binary classifier
 $Y \in \{\text{lion, dog}\}$
Multi-class classification

$D(X, Y)$
↑
(Real)

✓ (ii) Regression

(e.g.) $D(\underline{x})$

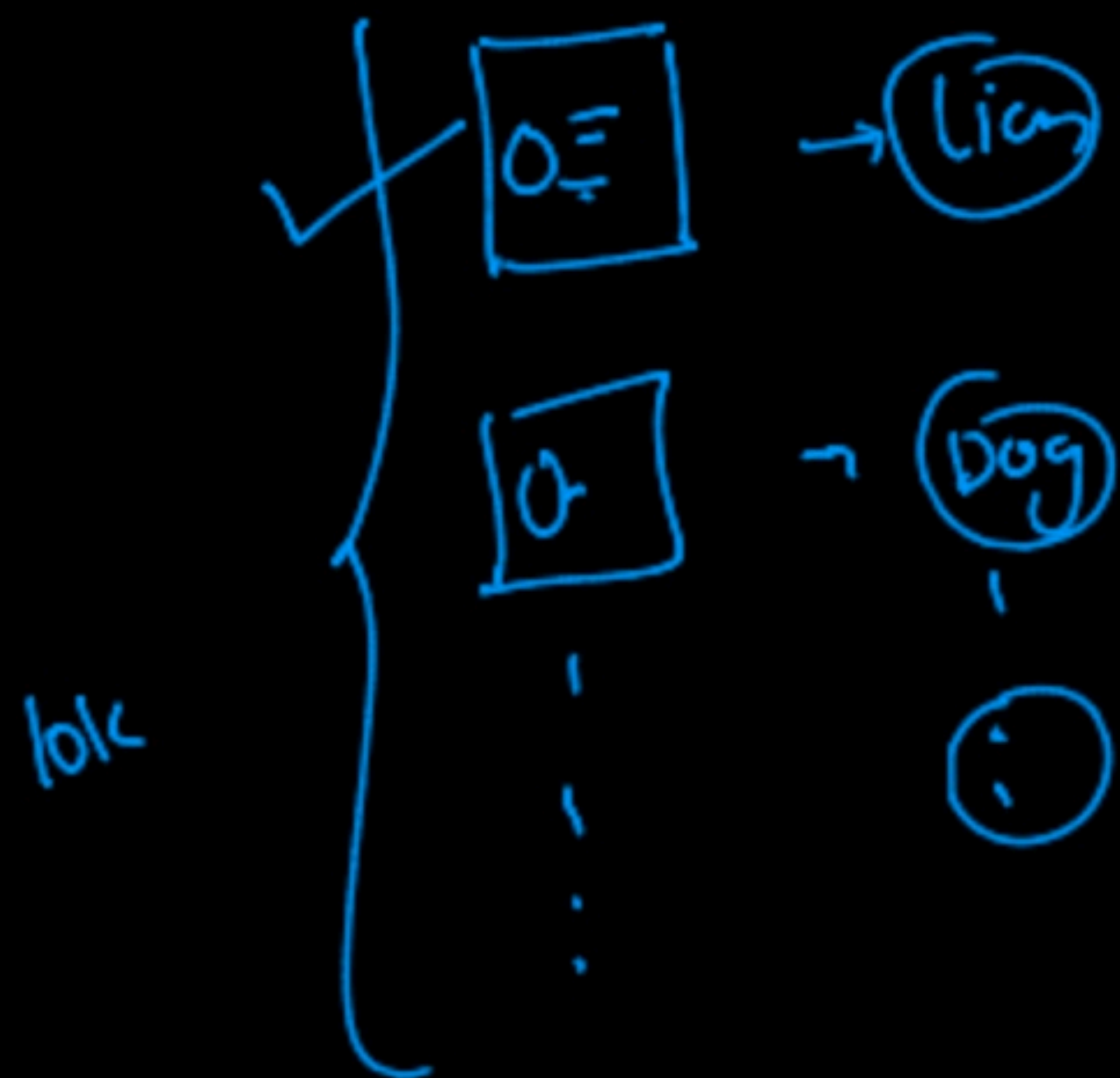
✓ Unsupervised training

(e.g.) (i) Clustering Algorithm
→ k-means ✓

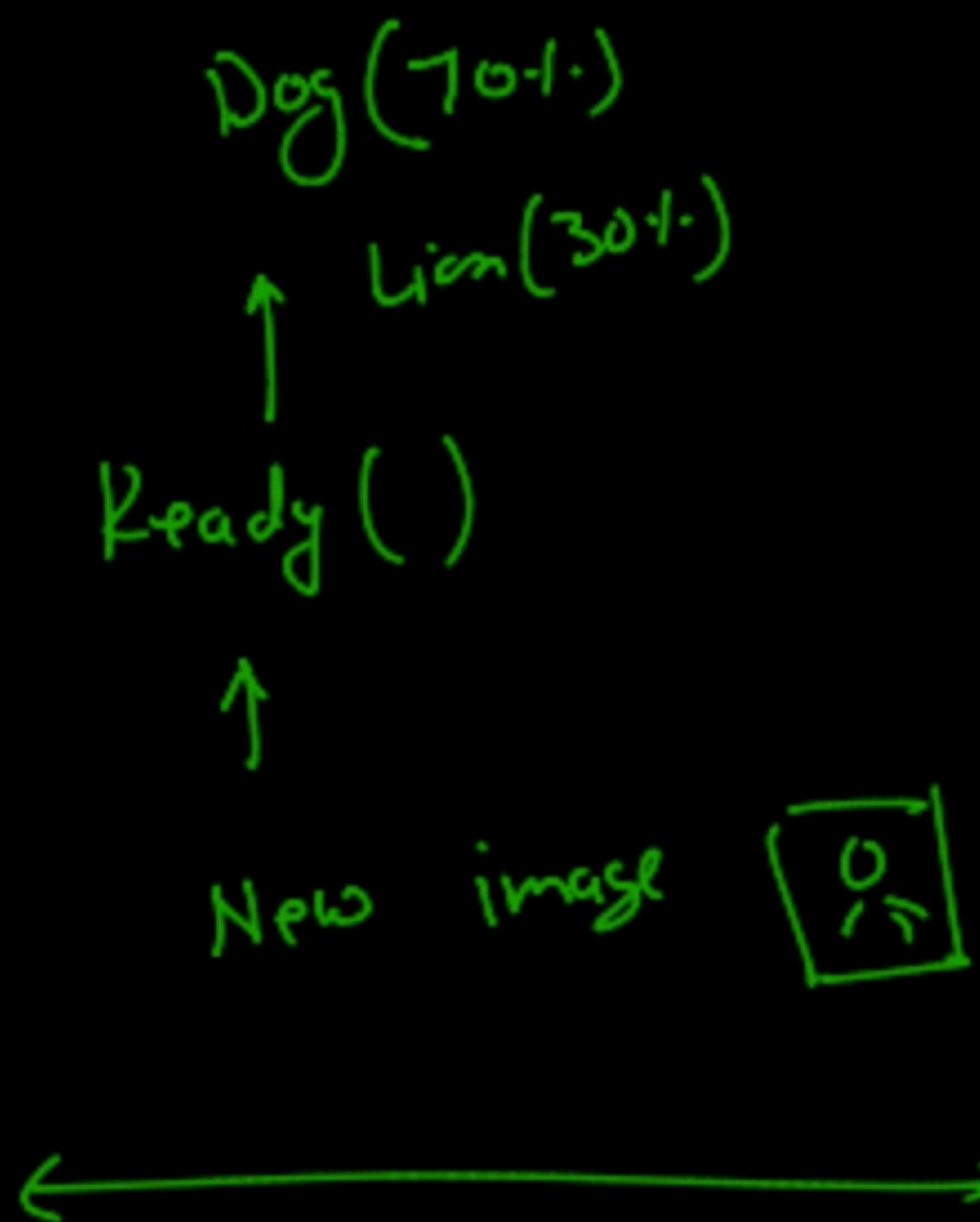
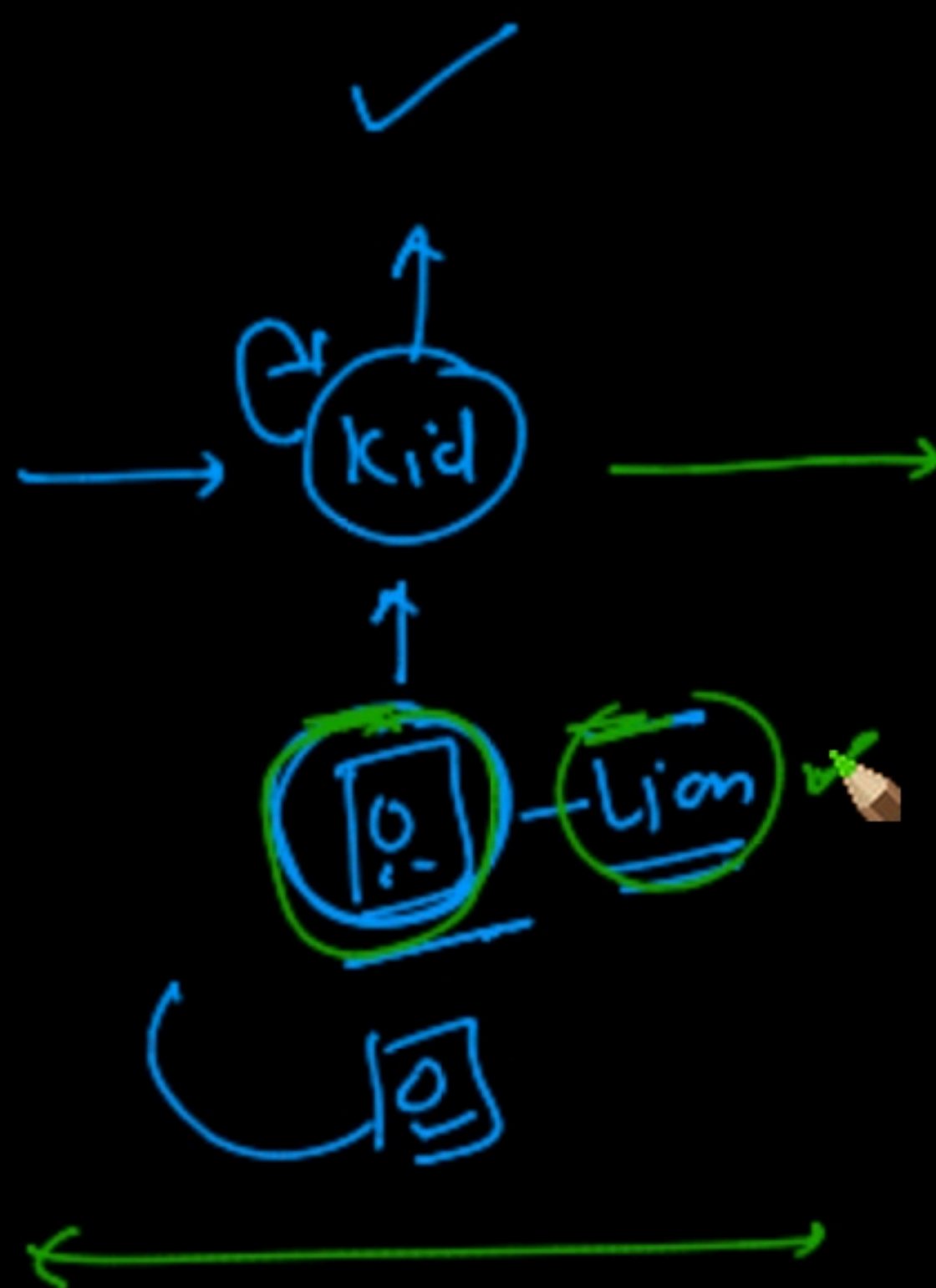
✓ (ii) PCA (Principal Component Analysis)

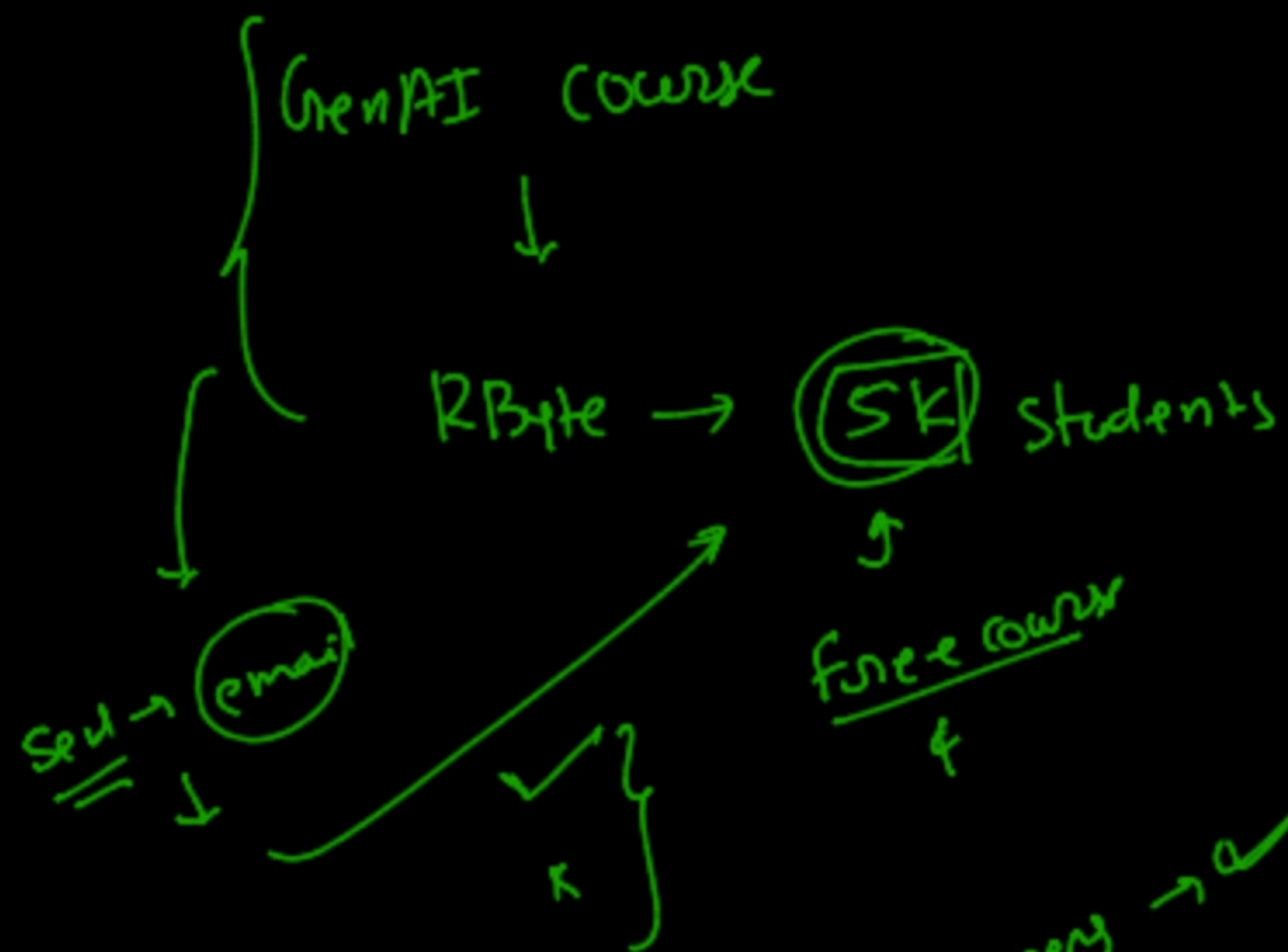
Example

eyes etc



(Lion/dog)





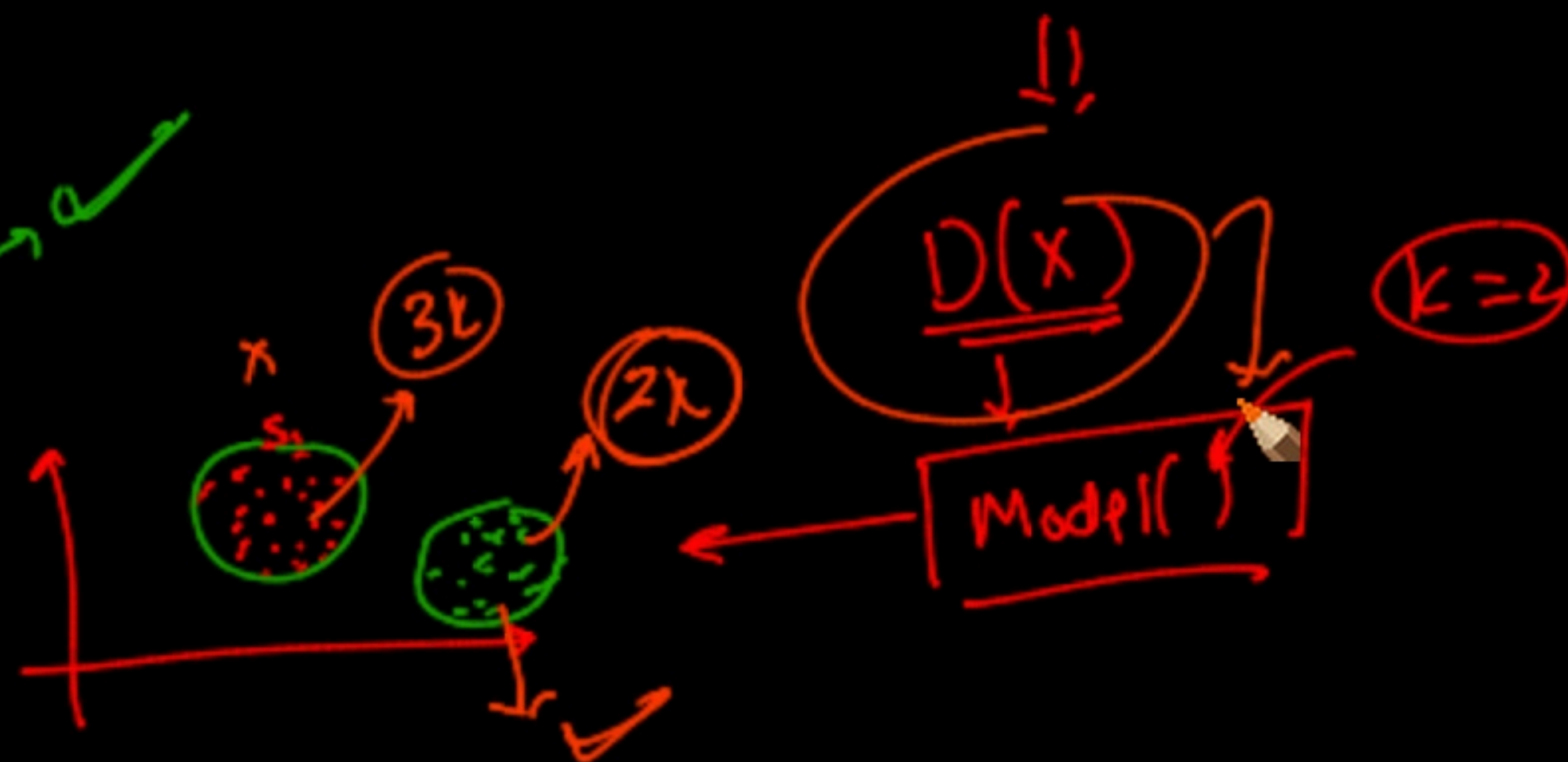
email ↓ cost ↑ → Best set of learners → a

Cost

X

	Augmented	log-w.	int.
S1	—	—	—
S2	—	—	—
S3			
⋮			
S ₅₀₀₀			

X



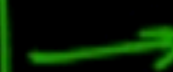
Eg.
=

Retail Domain
↑
(AI/ML/DS)

(A)

	X		Y
	discount	B/G1	sale-volume
Day 1	20%	F	350
Day 2	25%	T	400
⋮	⋮	⋮	⋮
Day 100	⋮	⋮	⋮

$D(x, y)$



Sale-volume → Key
400/500

↑
(18, F)

Patients Data

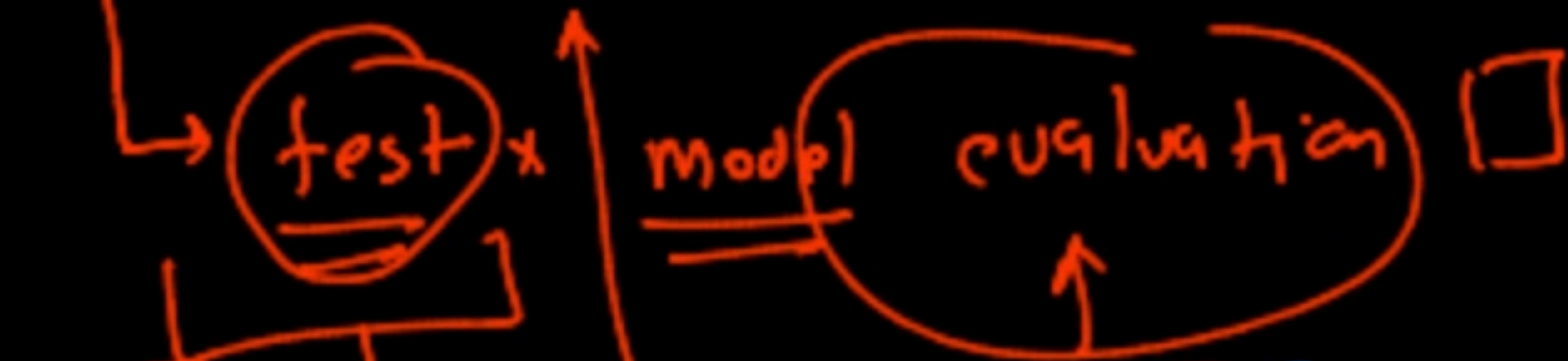
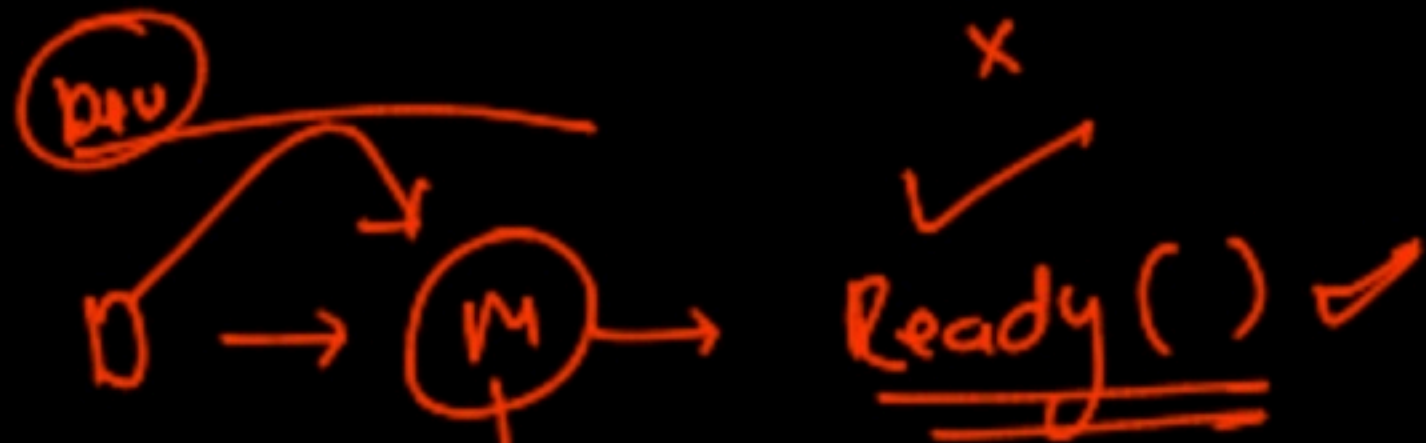
	X				Y
	BP	✓	✓	✓	Diast
P1					Type I
P2					Type II
⋮					⋮
P100					⋮

C [Type-I, II, III]

↓
 $m(X, Y) \rightarrow \text{Ready}() \rightarrow$
↑
[-----]

ST

- (i) classification
- (ii) regression



- (1) Acc
 - (2) Confusion Matrix
- FI
by me

eg.

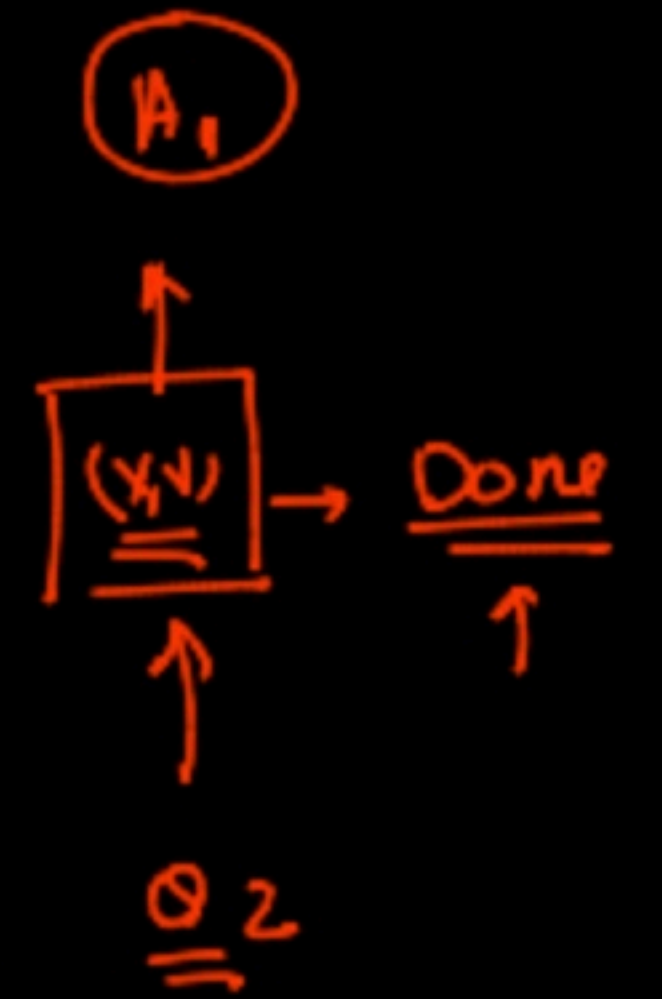
Exam
↓
Book(data)

$Y \in \{A_1, A_2\}$

$Q_1 - A_1$
$\vdots$
$Q_{1000} - A$

80% training

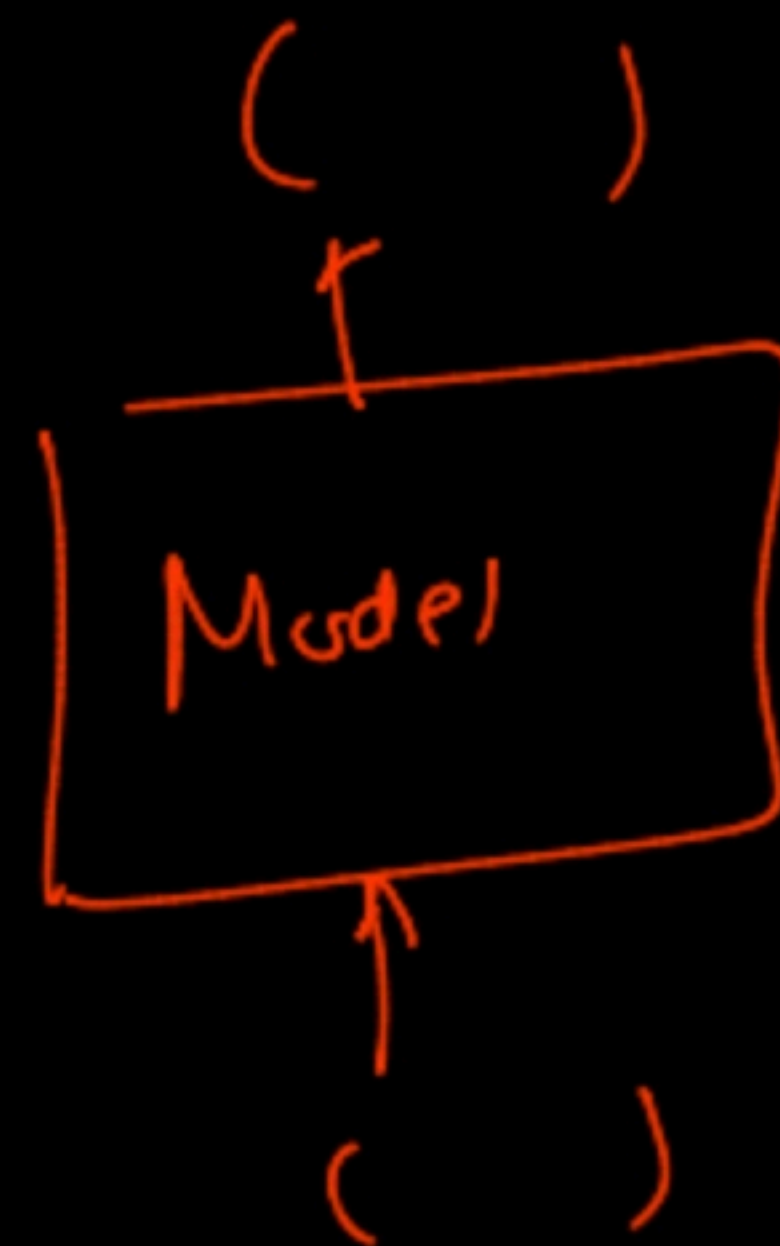
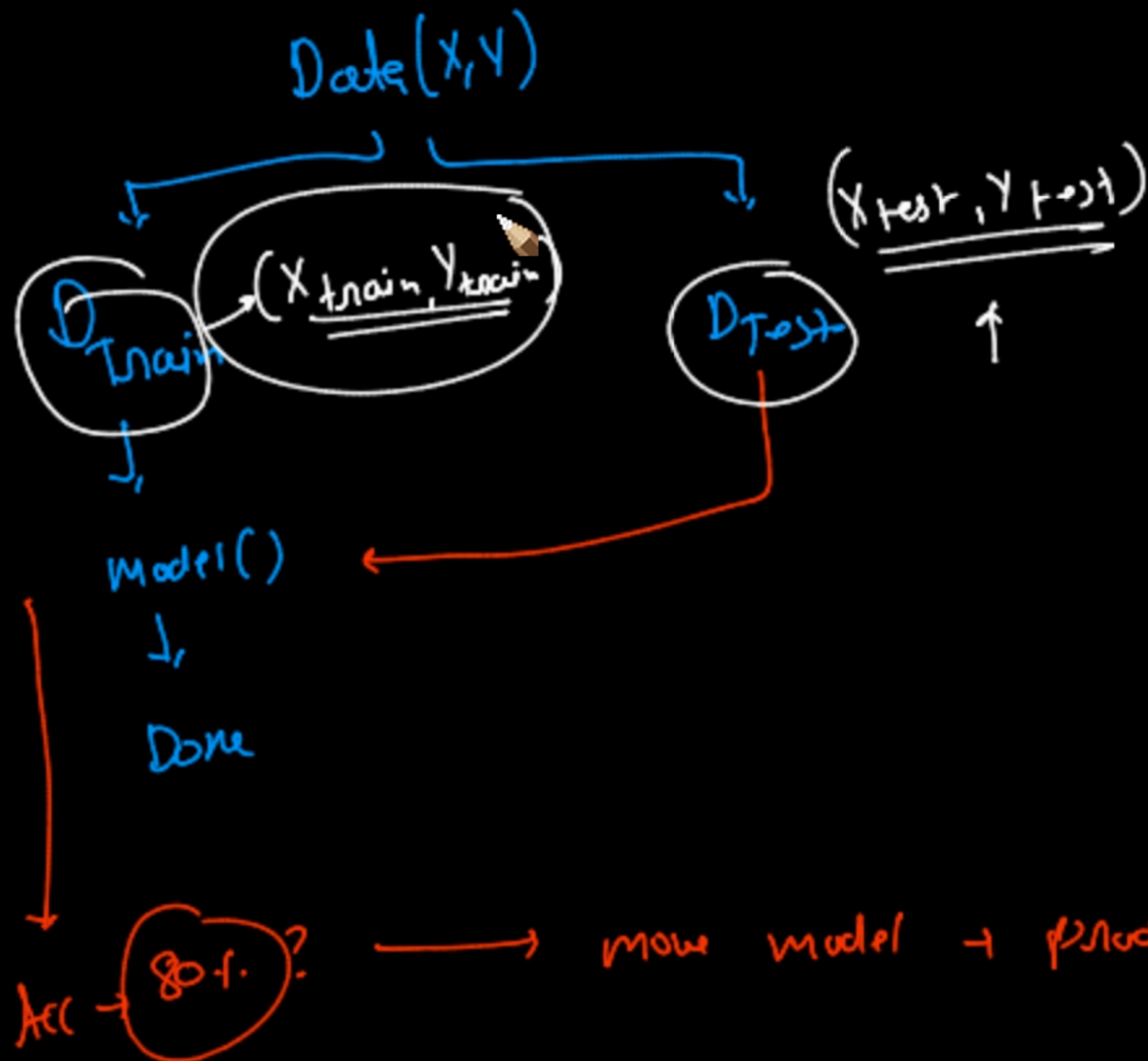
20% testing



Q	A
Q1	A1
Q2	A1
Q3	A2
Q4	A2

→ FI
by me

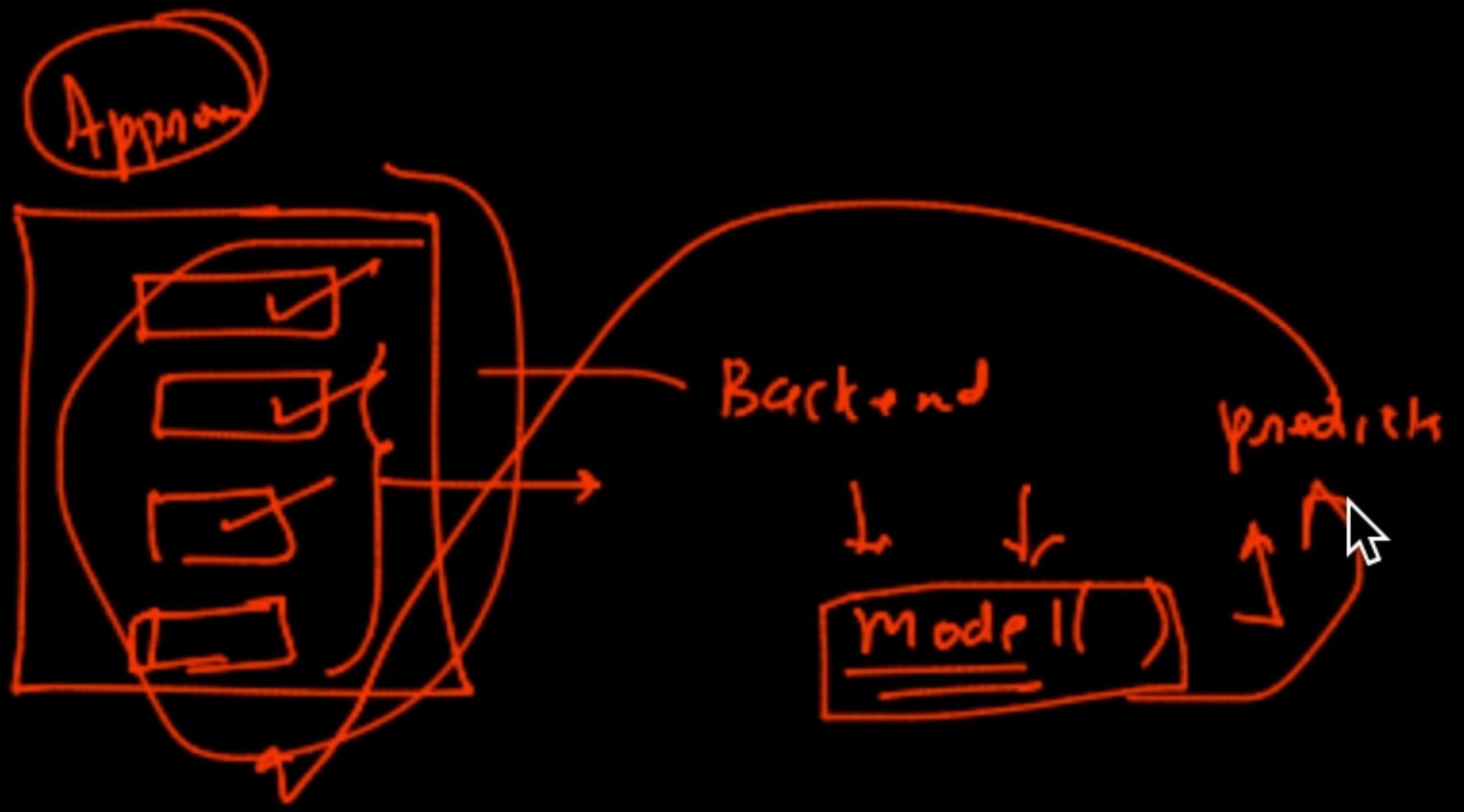
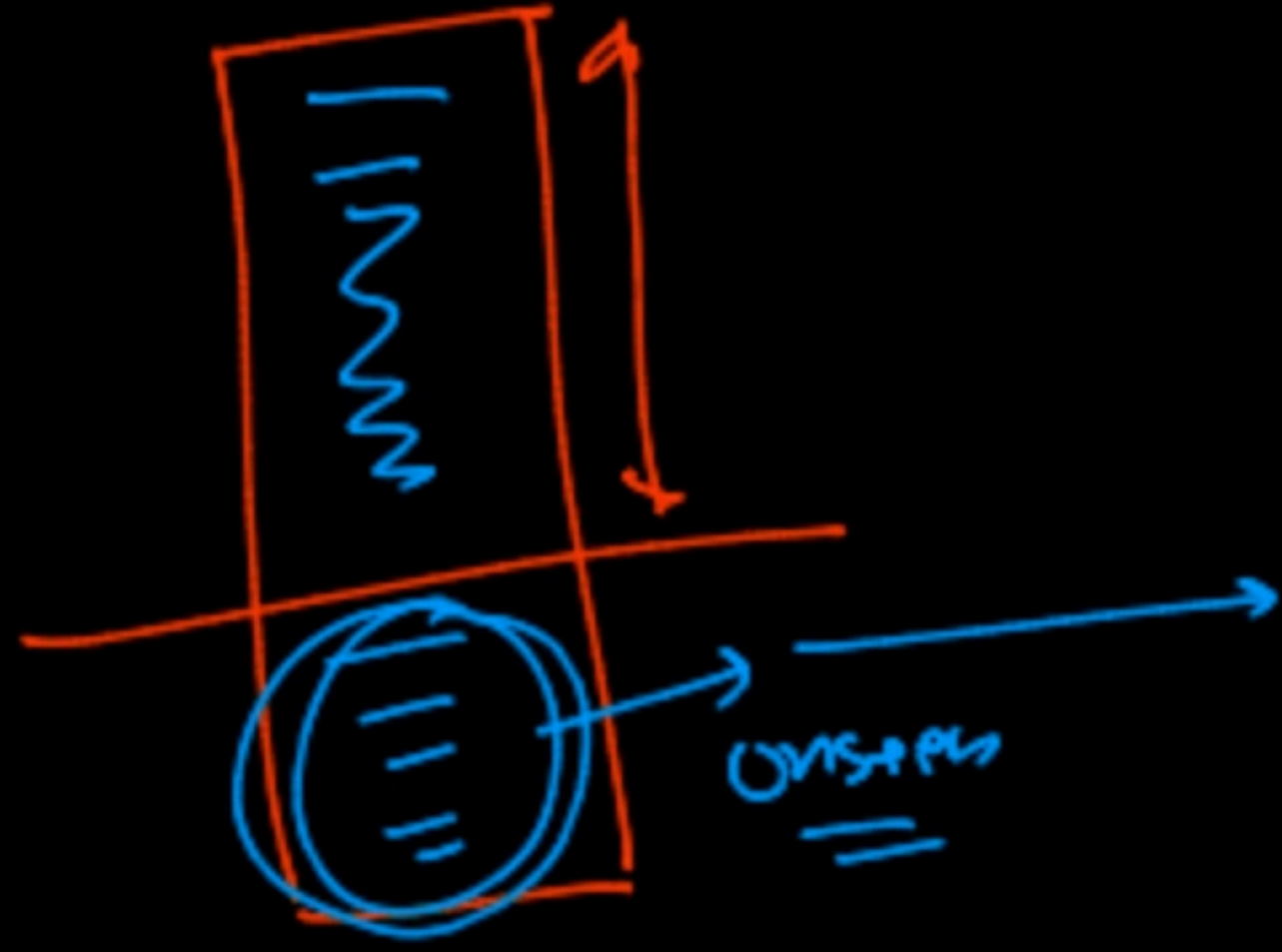
BA
 $\frac{2}{4}$
50%



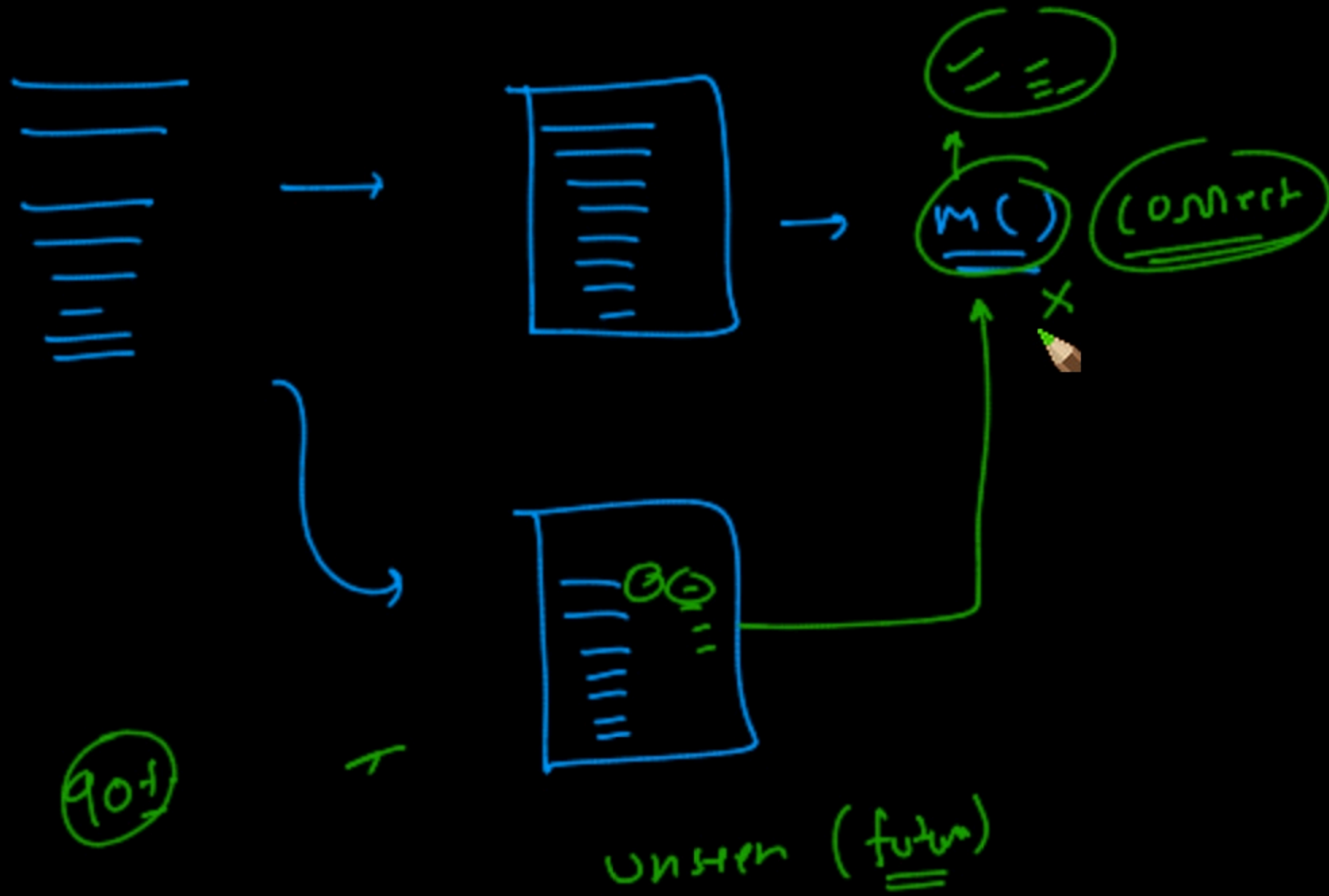
HDFC

↓
posita

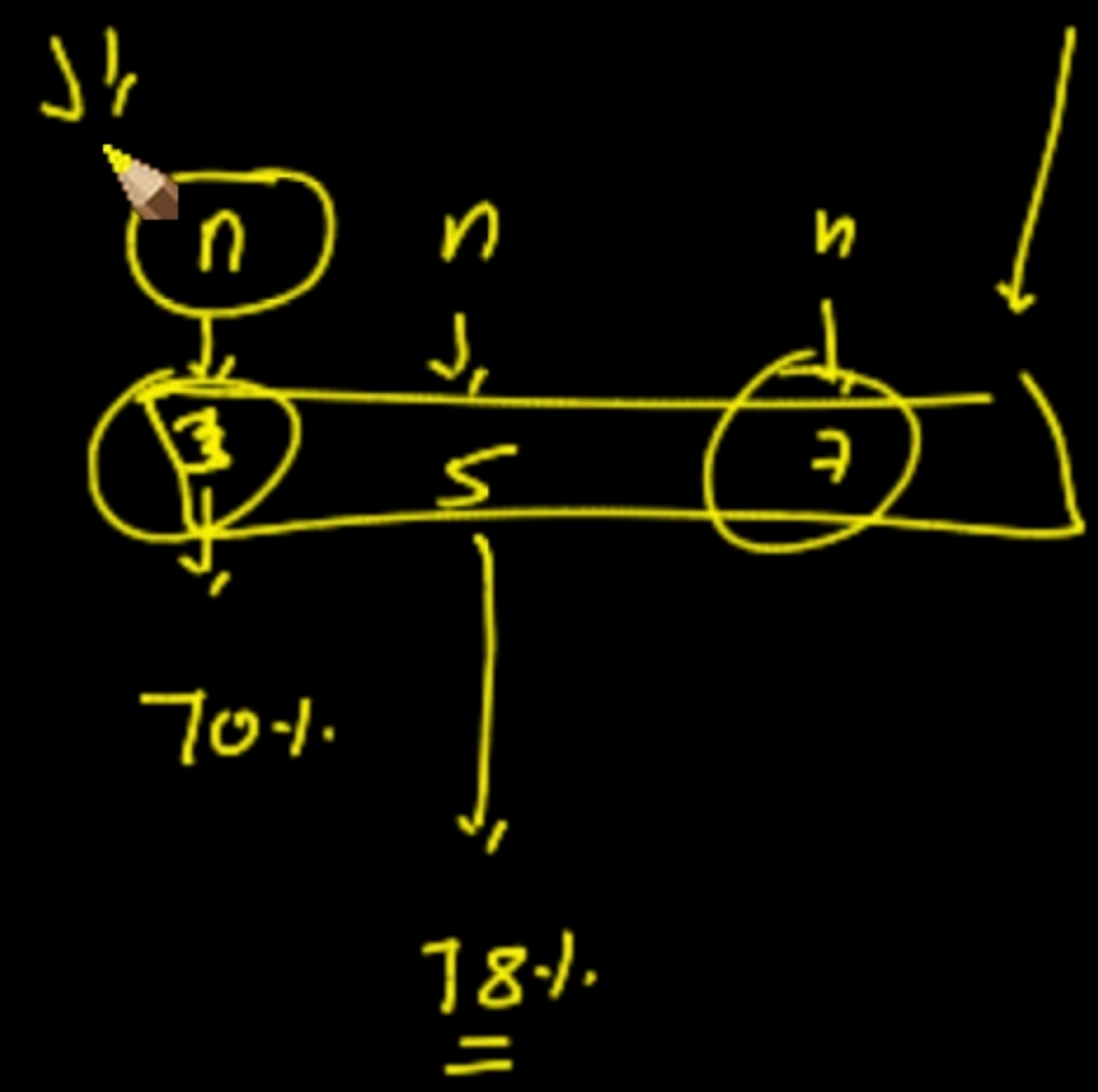
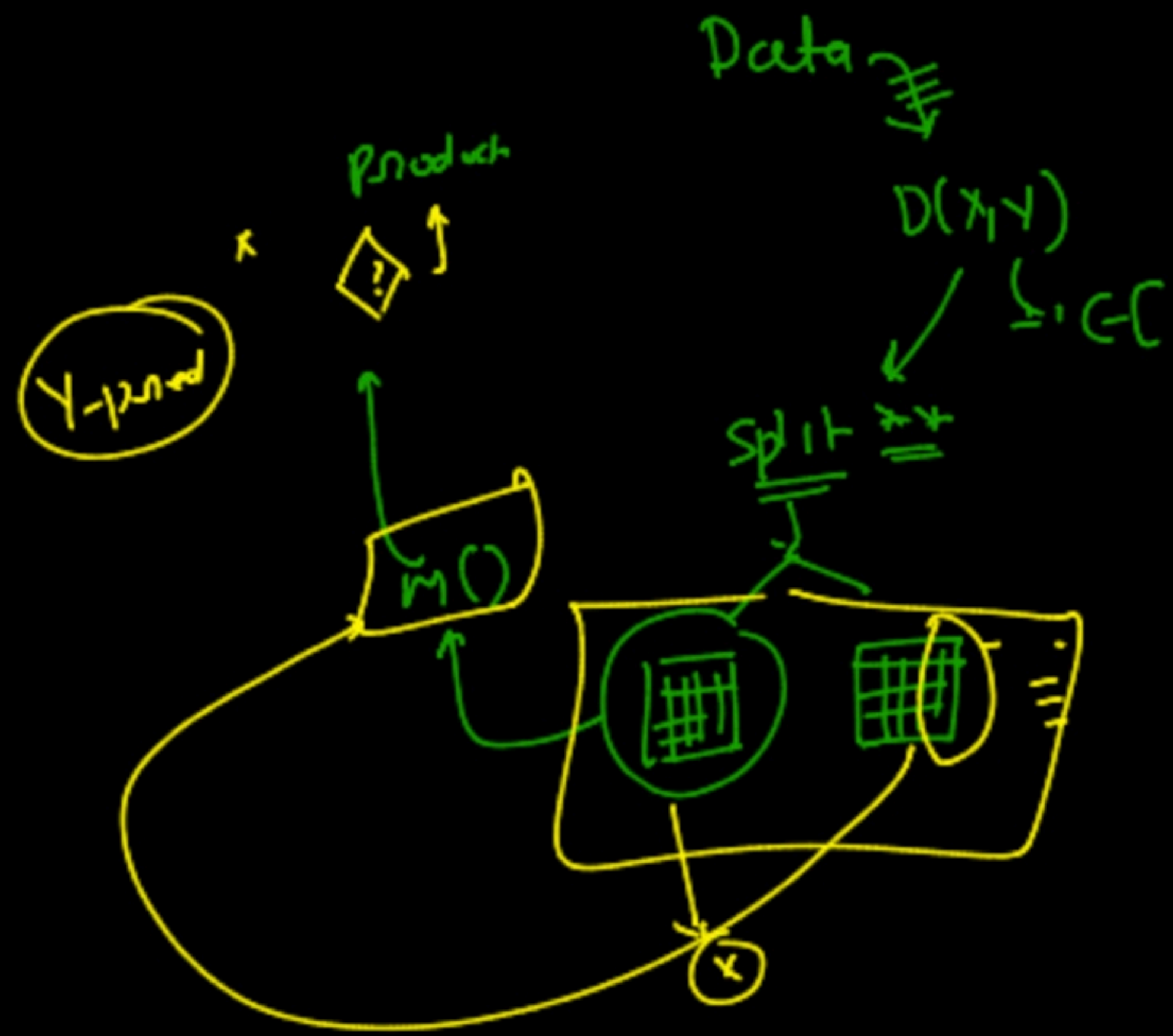
↓
loan →

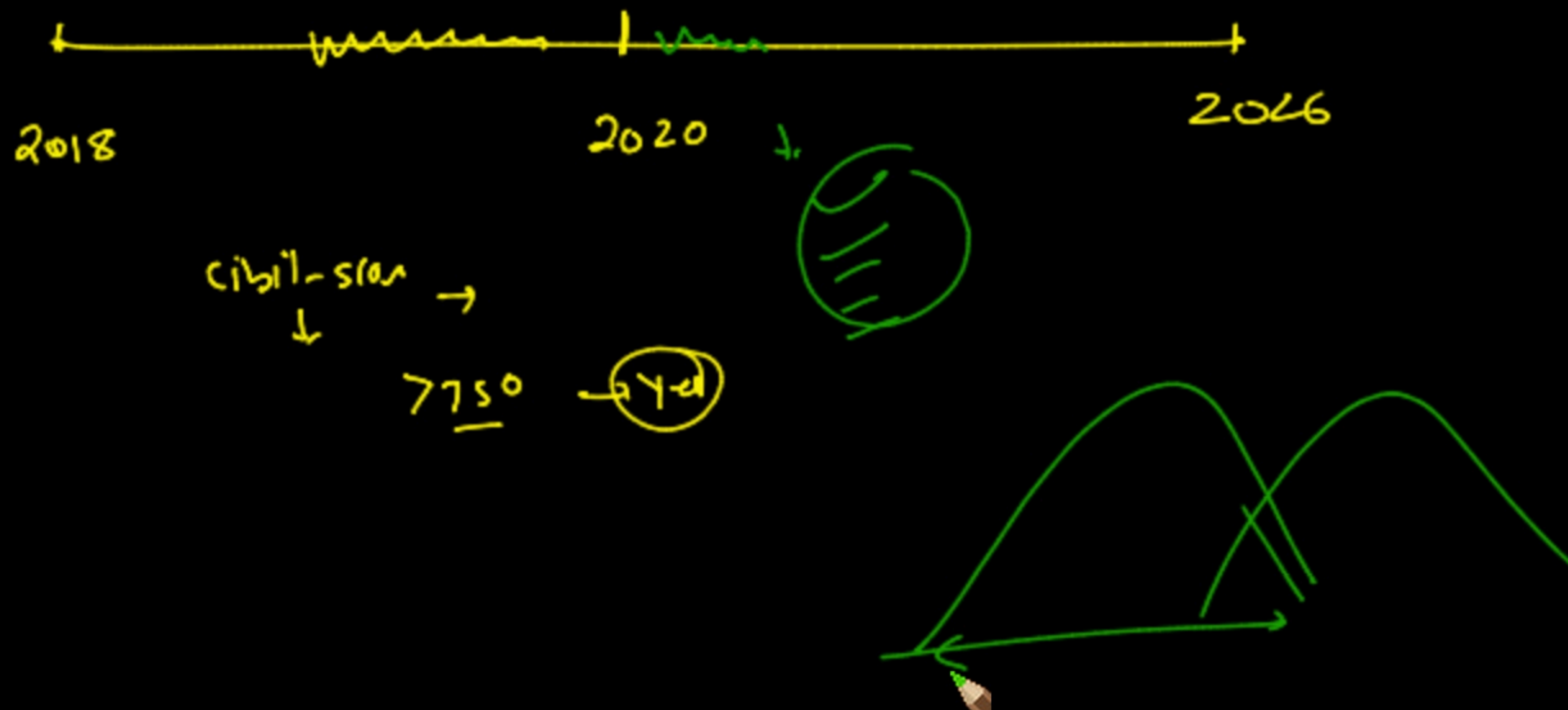


1000 points



70%





Distance based Approach

kNN Algorithm

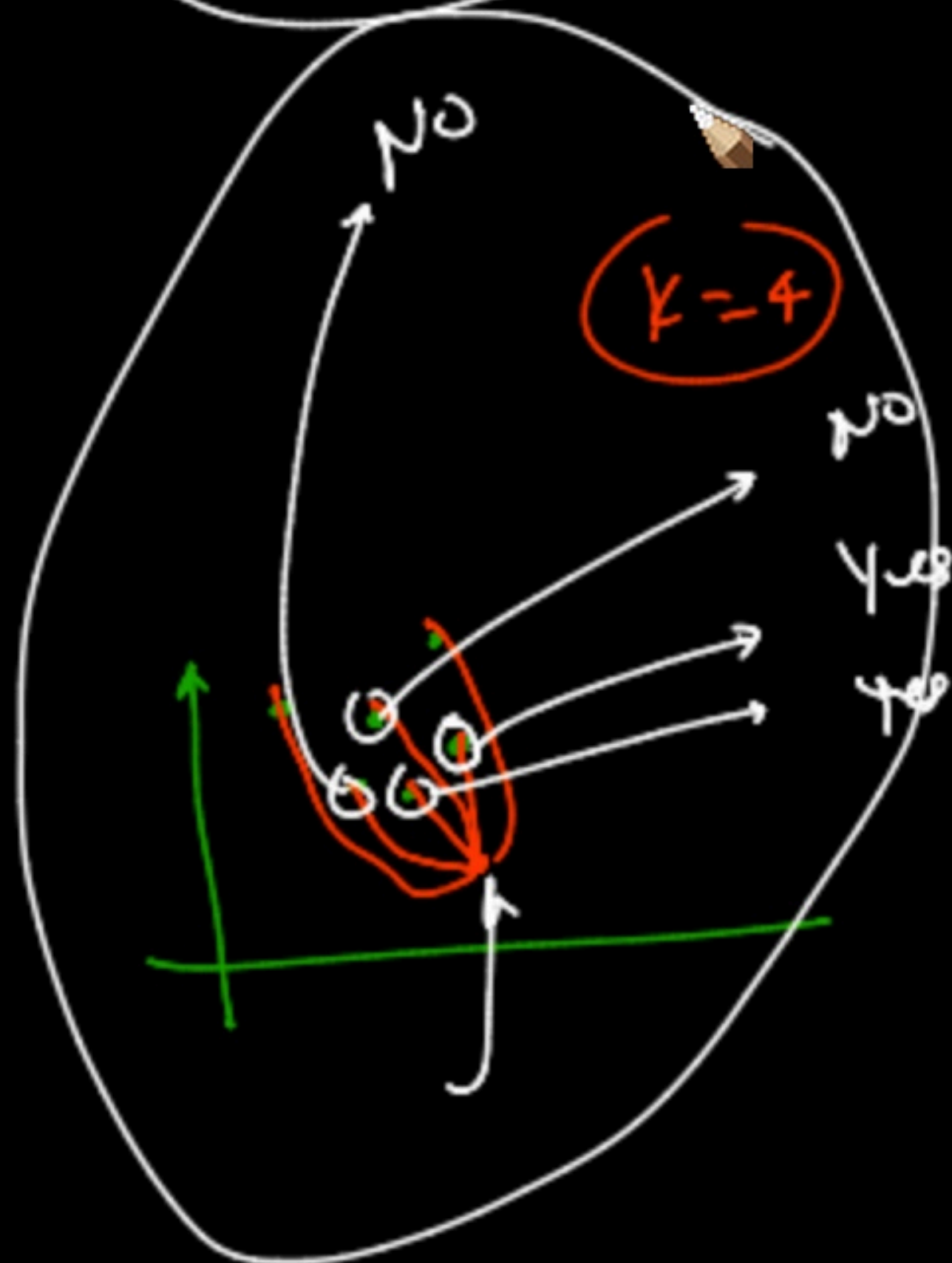
Sklearn

Int (odd)

$k=?$

odd number
classification

$k=4$



① distance

② sort it ($k \leq N$)

③ classification

→ majority voting

↳ $k=4$

Yes → 2
No → 2

sort sort